

Kernel width and support strength of the RBLapSum surrogate under the code residual

Research note: a (reach, strength) grid at $K + J = 512$

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1 Purpose and summary of conclusions

The K+J notes compared, on an 8-layer TinyStories model with an abs-TopK bottleneck after every block, the RBLapSum `through_rank_kappa` surrogate – K active features, gradient to a pool of $K + J$ – with hard Top- K' at $K' = K + J$. Every surrogate run so far used one kernel width, $T = 2$ (or $T = 1$ with the output norm), and the full support term, $\gamma = 1$. With the code carried between blocks (`docs/kj-vs-hard-topk-code-residual.tex`) the surrogate’s margin over hard Top- K shrank to 0.047 at $K = 32$ and reversed at $K \geq 128$, and the support-scale sweep of that note showed that the best γ depends on K . This note asks whether the code-carried surrogate improves when the kernel width and the strength of the support term are set relative to the score geometry of each token rather than fixed: the width as a multiple of the pool’s score span or of the boundary score, the strength as the ratio of the support gradient to the read gradient at the rank boundary.

The grid is $\tau \in \{0.25, 0.5, 1\}$ for the width, $s \in \{0.1, 0.25, 0.5, 1\}$ for the strength, two width rules, and the cells $(K, J) = (32, 480)$ and $(256, 256)$: 48 runs of 10k steps, compared at matched steps with the code-carried hard Top- K' ladder and with the $T = 2$, $\gamma = 1$ runs of the earlier grid. A second arm repeats the span rule with the first-order scope (Section 7).

The main conclusions are:

- **No cell beats the fixed configuration at $K = 32$.** The best cells of either rule end 0.046 below hard Top-32 at step 10000; the $T = 2$, $\gamma = 1$ run ends 0.049 below. Within the stable region the loss falls monotonically with s at every width and, under the span rule, with τ at every s . The best cells have a realized support scale $\gamma = 2sT/b$ of 0.5–0.6.
- **A realized scale $\gamma \gtrsim 1$ is unstable under the relative rules.** Every cell with $\gamma \geq 1$ shows an exponential growth of the boundary score: by a factor 1.5–1.6 per thousand steps after step 3000 in the span cell $(\tau, s) = (1, 1)$ at $K = 32$, to $b = 180$ at 10k, with the loss advantage decaying from 0.040 to 0.007 as it grows. The relative rules make the surrogate invariant to the score scale; the fixed configuration holds b near 2.3 throughout training.
- **At $K = 256$ the surrogate gains at most 0.005.** Two cells end at or below hard Top-256, the span cell $(0.5, 0.1)$ at -0.005 and the b -rule cell $(0.25, 0.25)$ at -0.001 , both within single-seed variation. Every stronger or wider cell loses, monotonically in s and in τ , and there the boundary score shrinks rather than grows. All 24 cells beat the $T = 2$, $\gamma = 1$ run (+0.031).
- **Both rules order the grid the same way**, and their best cells coincide in realized γ . The span rule is the more sensitive to τ at $K = 32$ (the loss keeps falling to $\tau = 1$ at $s \leq 0.5$) and the b rule reaches its best at the narrowest width.

2 Definitions

The support term. Bottleneck ℓ scores its $N = 1536$ features by $z = E_\ell x$ and keeps the K of largest magnitude; the sorted magnitudes are $s_{(1)} \geq s_{(2)} \geq \dots$, the boundary score is $b = s_{(K+1)}$ and the pool is the next J features, ranks $K + 1$ to $K + J$. In the backward pass the surrogate adds to the hard gradient a support term over the pool. With u_i the value of pool member i , g_i the gradient that reaches it from the read path, q_i and a_j the exchange quantities of the `through_rank_kappa` mode, it is

$$\gamma \operatorname{sign}(u_i) \kappa_i (g_i u_i - q_i \sum_j a_j), \quad \kappa_i = \frac{1}{2T} e^{-|s_i - b|/T}, \quad (1)$$

where T is the kernel width and γ the support scale (`rblapsum_support_scale`). The kernel is a Laplace density centred at the boundary, so a member at distance $|s_i - b| \gg T$ receives almost nothing; the number of pool members that take part is set by T relative to the spacing of pool scores (the *reach*), and the size of the term for a member at the boundary by $\gamma |u| \kappa(b) = \gamma b / 2T$ (the *strength*).

The two width rules. `rblapsum_kernel_width` selects, per token,

$$T = \tau (s_{(K+1)} - s_{(K+J)}) \quad (\text{relative_span}), \quad T = \tau s_{(K+1)} \quad (\text{relative_b}), \quad (2)$$

with τ supplied by the existing `temperature` field. Under the span rule $\tau = 1$ sets the width equal to the whole pool’s score range, so at $\tau \geq 1$ every candidate takes part and $\tau = 0.25$ reaches about the top quarter of the pool. The ratio b/span was measured on the checkpoint ladders of the $T = 2$ runs at 1.6 for $K = 32$ and 2.8 for $K = 256$, nearly constant over training, so the b rule is the span rule with τ multiplied by that ratio; the two rules differ in what they track when the pool’s spread and the boundary score move apart.

The strength. `rblapsum_support_strength = s` replaces the fixed scale by the per-token value

$$\gamma = \frac{2sT}{b}, \quad \text{so that} \quad \gamma b \kappa(b) = s : \quad (3)$$

the support gradient of the $(K+1)$ -th feature equals s times its read gradient g , whatever T and b are. In these terms the fixed configuration $T = 2$, $\gamma = 1$ has a strength that decays over training as b falls: $s = b/4 \approx 0.93$ at initialization, 0.5 at step 2000 and 0.22 at step 20000 on the ladders, with a width that already covers the whole pool at $K = 32$. The grid holds s fixed instead and lets γ follow T/b . In the pool scope the same term also acts along the carried code, and the earlier studies showed that this is where the surrogate and the carry conflict (`docs/carry-surrogate.tex`); the grid does not change the scope, so γ scales both the read and the carry action of the term.

Both rules together make the surrogate invariant to a common rescaling of the scores: T and γ follow b , and the term (1) is unchanged. Nothing then anchors the score scale, and Section 4.1 shows where this matters.

3 Setup

Each run is the code-carried $T = 2$ cell of the $K+J$ grid (`ma_cr_rbk_k<K>_j<J>_t2.yaml` in `configs/cr_kj/`: 8×768 , abs-TopK over $N = 1536$, `residual_out` placement, decoupled gains, no output norm, pool scope, $b_0 = 0$, batch 96×512 , cosine schedule over 20k steps with 500 warm-up steps, seed 1337) with four lines changed: `temperature =  $\tau$` , `rblapsum_kernel_width`, and `rblapsum_support_strength = s`, and the run name. The grid is

rule	τ	s	cells (K, J)
span, $T = \tau(s_{(K+1)} - s_{(K+J)})$	0.25, 0.5, 1	0.1, 0.25, 0.5, 1	(32, 480), (256, 256)
b , $T = \tau s_{(K+1)}$	0.25, 0.5, 1	0.1, 0.25, 0.5, 1	(32, 480), (256, 256)

48 runs, configs/cr_kj/tsweep/ma_ts_{span,b}_t<tau>_s<s>_k<K>_j<J>, each trained for 10k steps of the 20k schedule and stopped with `train_guard.py --stop-step 10000` (the schedule is not compressed; the learning rate at step 10000 is that of a 20k run), validation every 500 steps on 40 batches, no checkpoints. The runs were trained on madrid, one RTX 5090 each, eight in parallel, at 0.41 s per step.

Baselines. All comparisons are at matched steps against runs of the code-carried $K+J$ grid read from `docs/figures/kj-cr/data/curves_new.json`: the hard Top- K' ladder with the output norm, `ma_cr_hard_k<K'>_pnorm` for $K' = 32, 64, 128, 256, 512$, and the $T = 2$, $\gamma = 1$ pool-scope cells `ma_cr_rbk_k32_j480_t2` and `ma_cr_rbk_k256_j256_t2`, the configuration every earlier surrogate run used. At step 10000 the ladder reads

hard Top- K' , code carried	32	64	128	256	512
val CE at step 10000	1.6560	1.5928	1.5389	1.5110	1.4822

and the $T = 2$ cells at 1.6067 (−0.049 against hard Top-32) and 1.5424 (+0.031 against hard Top-256).

Diagnostics. Every run logs the realized width and scale, averaged over tokens and blocks (`rb_temperature_eff`, `rb_support_scale_eff`), and the boundary score b (`rb_boundary`); the tables below quote their values at step 10000. The curves and diagnostics are in `curves_tsweep.json` under `docs/figures/tsweep/data/`; the figures are made by `scripts/plot_tsweep.py`.

4 Results: pool scope

Validation CE at step 10000 minus the code-carried hard Top- K at the same step (negative: the cell is better than hard Top- K). Rows are τ , columns s . Bold marks the best cell of each table; a dagger marks the cells whose score scale ran away (Section 4.1).

span rule, τ	$K = 32, J = 480$ (hard Top-32: 1.6560)				$K = 256, J = 256$ (hard Top-256: 1.5110)			
	$s=0.1$	0.25	0.5	1	$s=0.1$	0.25	0.5	1
0.25	−0.005	−0.009	−0.014	−0.033	−0.001	+0.001	+0.002	+0.005
0.5	−0.005	−0.011	−0.027	−0.046	−0.005	0.000	+0.005	+0.017
1	−0.005	−0.013	−0.036	−0.007 [†]	+0.004	+0.005	+0.010	+0.029
$T=2, \gamma=1$	−0.049				+0.031			

b rule, τ	$K = 32, J = 480$				$K = 256, J = 256$			
	$s=0.1$	0.25	0.5	1	$s=0.1$	0.25	0.5	1
0.25	−0.003	−0.017	−0.020	−0.046	+0.001	−0.001	+0.008	+0.017
0.5	−0.005	−0.015	−0.034	−0.020 [†]	+0.004	+0.006	+0.016	+0.025
1	−0.002	−0.012	−0.018 [†]	+0.047 [†]	+0.006	+0.010	+0.024	+0.055

Under the b rule the realized scale is $\gamma = 2s\tau$ by construction: 0.05 to 0.5 in the first row, 0.1 to 1 in the second, 0.2 to 2 in the third. Under the span rule it is $\gamma = 2s\tau \text{span}/b$ and was measured at step 10000 as 0.33 s , 0.65 s and 1.2 s for $\tau = 0.25, 0.5, 1$ at $K = 32$ (the last from the one stable cell of that row, $s = 0.5$) and as 0.20 s , 0.41 s , 0.75 s at $K = 256$.

$K = 32$. In every row of both tables the loss falls with s up to the last stable cell. Under the span rule it also falls with τ at every $s \leq 0.5$, from -0.014 to -0.036 at $s = 0.5$; the stable cell of largest τ and s , $(0.5, 1)$, is the best at -0.046 . Under the b rule the narrowest width is the best at $s = 1$ (-0.046 at $\tau = 0.25$), and the wider widths at $s = 1$ are the runaway cells. The two best cells have realized scales 0.60 (span) and 0.50 (b) and end within 0.0002 of each other (1.6103 and 1.6101). Neither reaches the $T = 2$, $\gamma = 1$ run (1.6067). The span cell $(0.5, 1)$ ran 0.01 below that run between steps 1500 and 4000 and converged to it after step 6000 (Figure 1); the b cell $(0.25, 1)$ tracks it to within 0.005 from step 1500 on (Figure 3). At $s = 0.1$ all six cells end within 0.003 of one another, 0.002 – 0.005 below hard Top-32: at that strength the width does not matter.

$K = 256$. The loss rises with s in every row and with τ in every column, except at $s = 0.1$ under the span rule, where $\tau = 0.5$ is 0.004 better than $\tau = 0.25$ and $\tau = 1$ is 0.009 worse. The two cells at or below hard Top-256, -0.005 and -0.001 , are the weak-and-moderate corner; the strong-and-wide corner loses 0.029 (span) and 0.055 (b). Every span cell and every b cell with $\gamma \leq 1$ beats the $T = 2$, $\gamma = 1$ run, which ends $+0.031$ above hard Top-256.

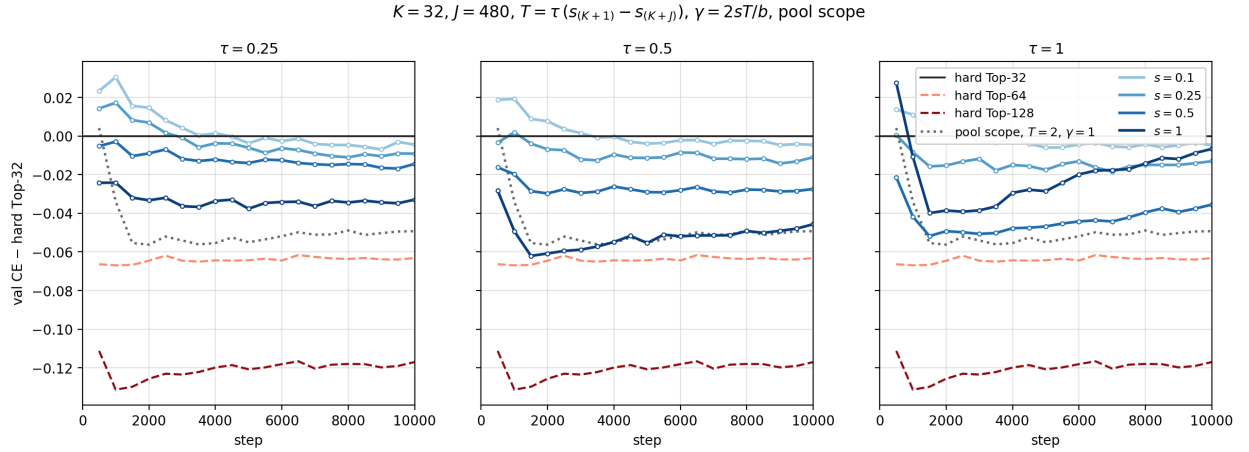


Figure 1: $K = 32$, $J = 480$, span rule, pool scope: validation CE minus code-carried hard Top-32, one panel per τ , the four strengths as a blue ramp light to dark with s . Dashed: hard Top-64 and Top-128 (reds); dotted grey: the $T = 2$, $\gamma = 1$ pool-scope run. The ordering in s holds in every panel; at $\tau = 1$ the $s = 1$ curve is the runaway cell, which leads until step 3000 and then climbs back towards hard Top-32 as its score scale grows.

4.1 The score scale

No run diverged in the sense of the guard (training CE more than 2.5 above its best for 800 steps), and the gradient norms stayed between 0.2 and 0.4 in every run. Four cells nevertheless left the regime of the others: the boundary score b , which falls slowly from about 2.6 at step 1000 to 0.8 – 1.5 at step 10000 in the weak cells, grows in them instead, and the cell’s loss advantage decays as it grows. The boundary score at step 10000 , averaged over tokens and blocks:

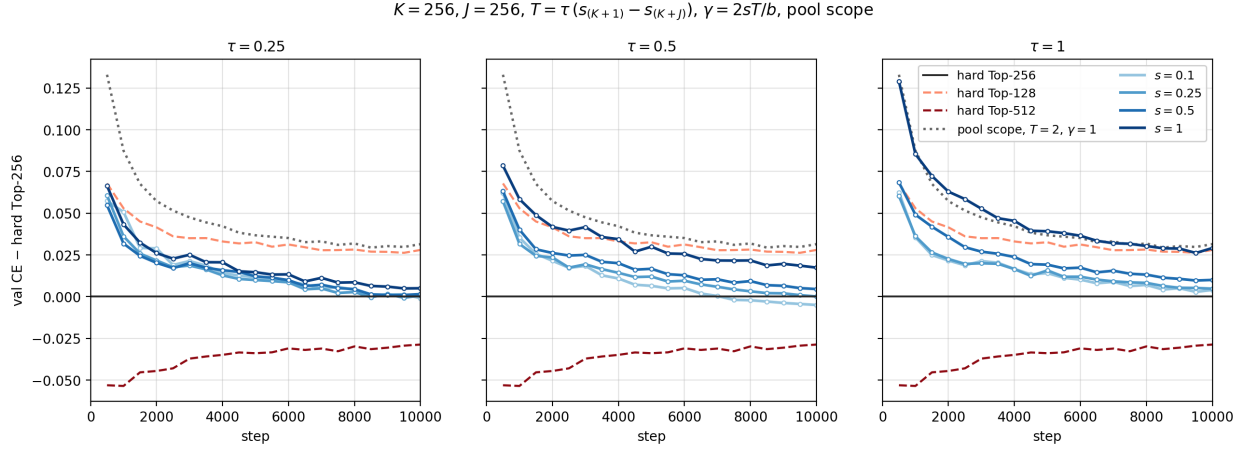


Figure 2: $K = 256, J = 256$, span rule, pool scope; same conventions, with hard Top-128 and hard Top-512 as the dashed references. The curves order by s in every panel, weakest lowest, and the whole family moves up with τ ; only the $(\tau, s) = (0.5, 0.1)$ curve crosses below hard Top-256, from step 7000 on. Every curve lies below the dotted $T = 2, \gamma = 1$ run from the first validation point except $(1, 1)$, which runs along it.

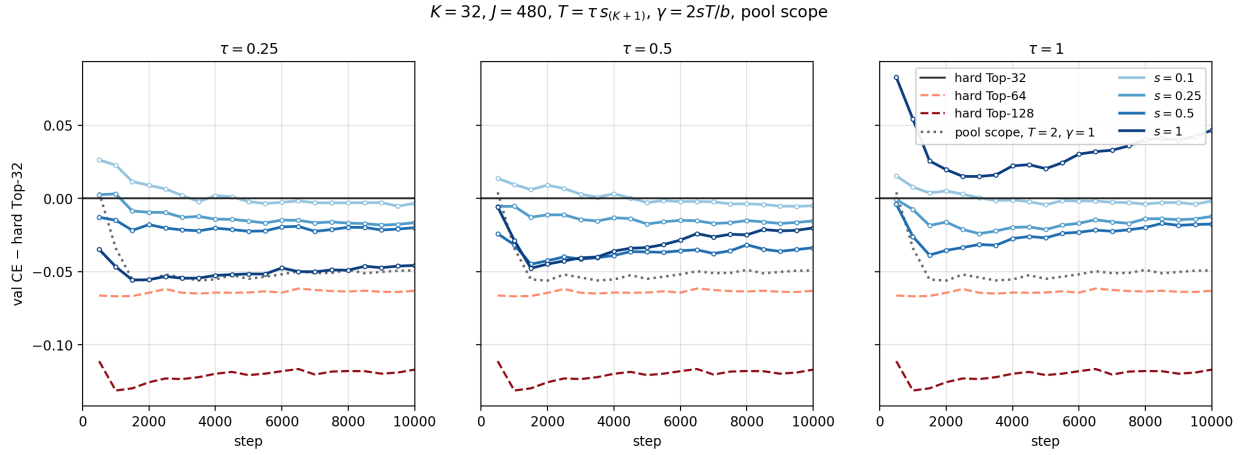


Figure 3: $K = 32, J = 480$, b rule, pool scope ($\gamma = 2s\tau$ in every cell). The $(\tau, s) = (0.25, 1)$ curve coincides with the $T = 2, \gamma = 1$ run from step 1500 on; the three runaway cells, $(0.5, 1)$, $(1, 0.5)$ and $(1, 1)$, all with $\gamma \geq 1$, turn upwards after their early lead, the last one crossing above hard Top-32 at step 4000 and ending 0.047 above it.

b at step 10000 τ	span rule, $K = 32$				b rule, $K = 32$			
	$s=0.1$	0.25	0.5	1	$s=0.1$	0.25	0.5	1
0.25	1.44	1.22	1.00	0.82	1.33	1.11	0.98	1.36
0.5	1.28	1.08	1.18	3.47	1.25	1.24	2.23	29.9
1	1.27	1.49	3.94	180	1.44	2.45	12.5	1561
	span rule, $K = 256$				b rule, $K = 256$			
0.25	1.16	1.11	1.11	1.01	1.05	0.93	0.76	0.63
0.5	1.11	1.00	0.85	0.72	0.92	0.74	0.63	1.62
1	0.94	0.76	0.64	0.66	0.93	0.92	1.38	3.43

At $K = 32$ the cells with a realized $\gamma \geq 1$ – span $(1, 1)$ with $\gamma = 1.1$, b -rule $(0.5, 1)$ and $(1, 0.5)$ with $\gamma = 1$, and $(1, 1)$ with $\gamma = 2$ – reach $b = 12$ to 1561; the cells at $\gamma = 0.5$ – 0.6 sit at 2.2–3.9, two to three times the weak cells; below that b is 0.8–1.5 as in the hard runs. In the span cell $(1, 1)$ the growth is exponential from step 3000 on:

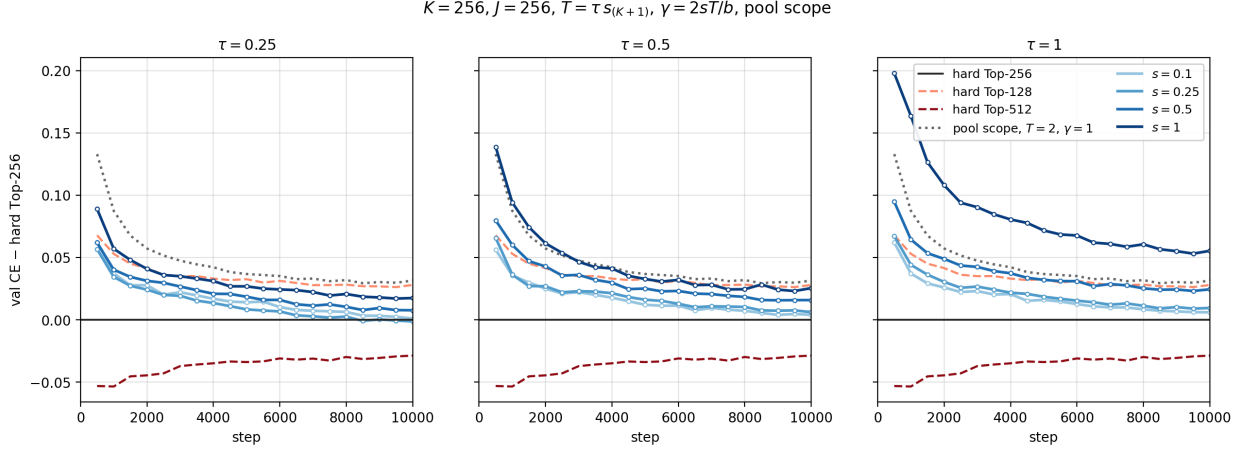


Figure 4: $K = 256, J = 256, b$ rule, pool scope. Monotone in s and in τ throughout; the $(0.25, 0.25)$ curve ends on hard Top-256 and the $(1, 1)$ curve, $\gamma = 2$, ends 0.055 above it, the worst cell of the grid.

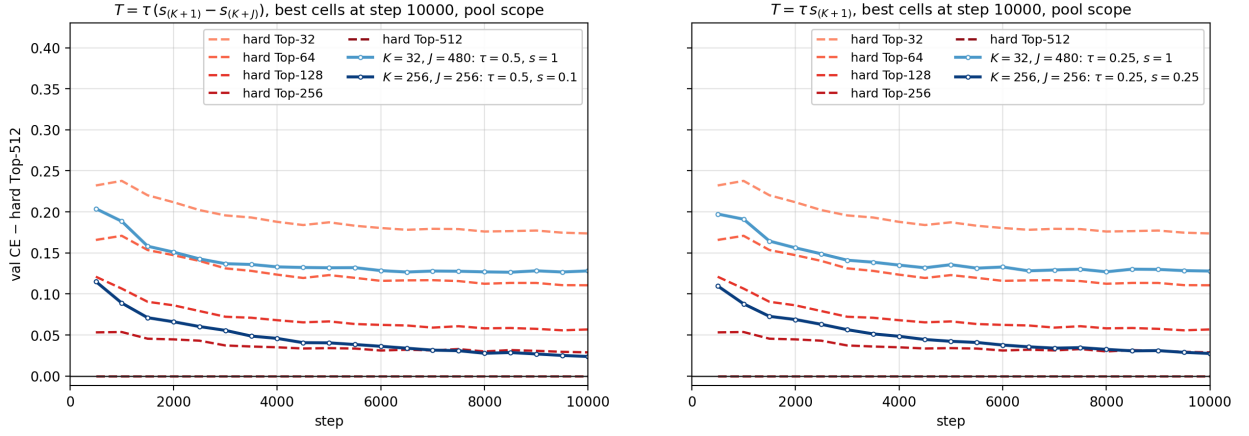


Figure 5: The best cell of each K at step 10000 against the code-carried hard ladder, validation CE minus hard Top-512; left the span rule, right the b rule. Dashed: hard Top- K' (reds light to dark with K'); solid: the best $K = 32$ cell (light blue) and the best $K = 256$ cell (dark blue). Both $K = 32$ cells end 0.017 above hard Top-64; both $K = 256$ cells end on hard Top-256, the span cell 0.005 below it.

step	1000	2000	3000	4000	5000	6000	7000	8000	10000
b	6.1	7.9	10.6	14.2	20.8	31.6	51.4	81.0	180.3
Δ to hard Top-32	-0.011	-0.039	-0.039	-0.029	-0.029	-0.020	-0.018	-0.014	-0.007

a factor 1.5–1.6 per thousand steps, while the realized γ stays at 1.1–1.2 and the realized T grows with b (to 121). The cell led hard Top-32 by 0.039 at steps 2000–3000, as much as the best cells, and gave up five sixths of it by step 10000. The b -rule cell $(1, 1)$ did the same faster and ended above hard Top-32. The fixed configuration, $T = 2, \gamma = 1$, keeps b near 2.3 to the end of a 20k run.

At $K = 256$ the strong cells move the other way: b falls to 0.63–0.66 at $(\tau, s) = (1, 1)$ under the span rule and at $(0.25, 1), (0.5, 0.5)$ under the b rule, against 1.0–1.2 in the weak cells; only the two b -rule cells with $\gamma \geq 1$ grow, to 1.6 and 3.4.

5 Interpretation

Three readings follow from the measurements; the first two are supported by the grid, the third is a hypothesis.

First, **the grid recovers the fixed configuration but does not improve on it at $K = 32$** . The best cells of both rules end within 0.004 of the $T = 2$, $\gamma = 1$ run and have a realized scale of 0.5–0.6, where that run’s realized strength $s = b/4$ is 0.2–0.5 over most of training with a width covering the pool. The loss falls monotonically towards the strong-and-wide corner and stops where the score scale becomes unstable, so the fixed configuration already sits near the optimum of this family, and what limits the code-carried surrogate at $K = 32$ is not the choice of reach or strength.

Second, **the relative rules remove the anchor of the score scale**. With $T \propto b$ (or $\propto$ span, which tracks b) and $\gamma \propto T/b$ the support term (1) is unchanged by a common rescaling of the scores, and the four cells with $\gamma \geq 1$ show that the gradient then drives the scale upwards without bound. At fixed T a growing b narrows the kernel relative to the pool and increases κ at the boundary only as $1/2T$, which is what keeps the $T = 2$ run at $b \approx 2.3$. A relative rule that is to be used at $\gamma \geq 1$ would need an explicit anchor, for instance a fixed T for the kernel together with a relative γ , or a penalty on b ; the grid did not test one.

Third, **the K -dependence**. At $K = 32$ more strength and more reach help; at $K = 256$ both hurt and the best cells are near zero. The hard ladder gives the scale of what a better support can be worth: the gain per added feature at the margin is 2.0×10^{-3} between Top-32 and Top-64 and 1.1×10^{-4} between Top-256 and Top-512, an 18-fold drop, while the costs of the term – the bias it puts on the gradient of the active features and its action along the carry – are set by s and τ and do not shrink with K . The direction matches; the size does not entirely: the $T = 2$ run recovers 0.049 of the 0.063 doubling gain at $K = 32$, and the same fraction of the 0.029 gain at $K = 256$ would be 0.023, where the best cells recover 0.005. The remainder is what the carry studies attributed to the term’s action along the carried code, which the pool scope keeps in every cell here.

6 Scope

All cells are single runs at one seed, and the comparisons are at step 10000 of a 20k schedule, not at convergence; in the earlier studies the ranking of methods moved by up to a quarter of the gaps between 3k and 10k steps. Differences below about 0.005 are within what one config produces on two machines. The hard baselines carry the output norm and the surrogate cells do not, as in the K+J notes. The grid changes the kernel width and the scale and keeps everything else of the code-carried $T = 2$ cell, including the pool scope, so the term’s action along the carry is present in every cell. Only two (K, J) cells were run, both at $K + J = 512$.

7 The first-order arm

The span rule of the grid, $\tau \in \{0.25, 0.5, 1\}$, $s \in \{0.1, 0.25, 0.5, 1\}$, at $(32, 480)$ and $(256, 256)$, repeated with the scope `rblapsum_surrogate_scope = first_order`: each gate’s support term is evaluated on the gradient that reached it along hard paths, so no gradient carries more than one support term and the products of terms along the carry are removed (`docs/kj-vs-hard-topk-code-residual.tex`, Section 6). The reference of this arm is the first-order $T = 2$, $\gamma = 1$ cell of the K+J grid, `ma_cr_rbk_k<K>_j<J>_t2_fo`, at 1.6322 (−0.024 against hard Top-32) and 1.5285 (+0.018 against hard Top-256) at step 10000. The 24 runs (`ma_ts_span_*_fo`) are training at the time of writing; this section is completed when they finish.